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Dark matter particle properties from galaxy rotation curves and the theory of energy cascade

Zhijie (Jay) Xu,[1]⋆
[1]*Physical and Computational Sciences Directorate, Pacific Northwest National Laboratory, Richland, WA 99354, USA*

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ABSTRACT

Collisionless dark matter passes kinetic energy from small haloes to large ones at a rate ε_u that does not depend on scale, much as turbulence passes energy between eddies. We measure this rate, test the scaling laws it implies, and follow the consequences down to the scale of a single particle. The cosmic energy equation fixes the potential-to-kinetic energy ratio at $7/5$ in the matter era; the rate is set by the growth of the kinetic energy of halo particles, and the kinetic energy of all particles, in and out of haloes, gives a convenient estimate, $\varepsilon_u = -(3/2)u_0^2/t_0 = -4.58\times10^{-7}\,\mathrm{m^2 s^{-3}}$ in the Virgo simulation. In Illustris-1-Dark this all-particle estimate falls from 7.9 to $3.3\times10^{-7}\,\mathrm{m^2 s^{-3}}$ between $z=12$ and $z=0$ and approaches the halo-particle rate, $2.6\times10^{-7}\,\mathrm{m^2 s^{-3}}$, as the fraction of mass in haloes grows to 55 per cent. A constant rate implies a two-thirds law for velocities and a four-thirds law, $\rho_r \propto |\varepsilon_u|^{2/3} G^{-1} r^{-4/3}$, for the mean density within r. The 409 rotation-curve haloes of SPARC, the DiskMass Survey and Sofue's compilation follow it over 3.9 decades in radius with a pooled slope of -1.37 (95 per cent interval -1.44 to -1.30), although single samples range from -1.07 to -1.77. If gravity is the only interaction, nothing stops the cascade above the scale where quantum effects set in. Combining the uncertainty principle, the virial theorem and the constant energy flux for two particles then gives a mass $m_X = (256|\varepsilon_u|\hbar^5/G^4)^{1/9} = 0.91\times10^{12}\,\mathrm{GeV}$, a size $l_X = 3.13\times10^{-13}\,\mathrm{m}$ and an energy scale $\hbar/t_X = 0.87\,\mathrm{neV}$. Because $m_X \propto |\varepsilon_u|^{1/9}$, the full range of rates allowed by the data moves the mass only between 0.82 and $0.99\times10^{12}\,\mathrm{GeV}$. The same mass follows independently from free streaming. A self-interaction of $1\,\mathrm{cm^2 g^{-1}}$ would dominate the cascade only below 0.046 pc.

Key words: dark matter – galaxies: haloes – galaxies: kinematics and dynamics – methods: numerical – cosmology: theory

CONTENTS

1 Introduction
2 The rate of energy cascade
  2.1 Cosmic energy equation
  2.2 What the simulations show
3 Scaling laws of a constant energy cascade
  3.1 Two-thirds, five-thirds and four-thirds laws
  3.2 Haloes in Illustris-1-Dark
  3.3 Galaxy rotation curves
4 Extension to the particle scale
  4.1 Three conditions for two particles
  4.2 How robust are the numbers?
  4.3 Comparison with other routes
5 Self-interacting dark matter
6 Conclusions

1 INTRODUCTION

Dark matter makes up about five-sixths of the matter in the Universe. Its presence is established by the flat rotation curves of spiral galaxies [1, 2] and its abundance is fixed precisely by the anisotropies of the cosmic microwave background [3], but both measure only its gravity. The mass of the dark matter particle is unknown. Candidates span many orders of magnitude, from light axions [4] through thermal relics of weak-scale mass [5, 6] to superheavy particles of 10^{12}–$10^{16}\,\mathrm{GeV}$ produced gravitationally at the end of inflation [7, 8, 9]. Weak-scale relics, the long-favoured candidates, have not been detected. On galactic scales, the cusp–core and related small-scale problems of cold dark matter [10, 11] have motivated warm [12] and self-interacting [13, 14] alternatives.

Astrophysical data constrain the particle mass mainly through small-scale structure. The Lyman-α forest bounds the mass of warm dark matter from below at a few keV [12], and the densities of dwarf galaxies and clusters bound the self-interaction cross-section per unit mass [14]. Both are limits on departures from cold, collisionless behaviour. For dark matter that is cold and interacts only through gravity, the particle mass leaves no imprint on the structures that can be observed, and in most models it is an input, chosen by a production mechanism rather than derived from the dynamics of dark matter itself.

The dynamics of dark matter on non-linear scales bear a close analogy to turbulence. In a turbulent fluid, kinetic energy is passed from large to small eddies at a rate ε that does not depend on scale [15, 16], and in two dimensions it is passed the other way, from small to large scales [17]. Self-gravitating collisionless dark matter

⋆ E-mail: zhijie.xu@pnnl.gov; zhijiexu@hotmail.com

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is likewise random, non-linear and multiscale, but its structures are haloes rather than eddies, and haloes grow by merging. Mass and kinetic energy are therefore carried from small to large mass scales in an inverse cascade (Fig. 1), at rates that are independent of halo mass below a characteristic mass [18, 19, 20]. The rate of the energy cascade, ε_u, is set by the growth of the kinetic energy of halo particles. A random walk in scale space reproduces both the Kolmogorov spectrum of turbulence and the mass functions and density profiles of haloes [21], and a constant ε_u implies scaling laws for the velocity, mass and density of haloes that can be tested against galaxy rotation curves [22].

The analogy also raises a question that has no counterpart in fluids. In turbulence the cascade ends at the Kolmogorov scale, where viscosity converts the kinetic energy into heat. Collisionless dark matter has no viscosity. If gravity is its only interaction, nothing ends the cascade until quantum effects set in, at the scale of the particles themselves. The particles must then be compatible with the rate at which the cascade reaches them: the rate ε_u, together with $\hbar$ and G, should determine their mass, size and a characteristic energy. A related argument, which compares the free-streaming mass with the particle mass, gives $m_X \approx 10^{12}\,\mathrm{GeV}$ and is the basis of a cold freeze-out of superheavy dark matter [23, 24]; those papers treat the production, relic abundance, Hubble tension and cosmic-ray signals of such particles.

In this paper we follow the cascade route alone and ask how well the data determine its ingredients and conclusions. We measure ε_u in two cosmological simulations, using the kinetic energy of all particles as a convenient estimate and that of halo particles as the definition, and show how the two approach each other as haloes come to dominate the mass. We test the scaling laws implied by a constant rate against 409 haloes from three samples of decomposed rotation curves, sample by sample. We then extend the cascade to the particle scale, obtain $m_X = 0.91 \times 10^{12}\,\mathrm{GeV}$, and show that the full range of rates allowed by the data changes it only to 0.82–$0.99 \times 10^{12}\,\mathrm{GeV}$. Finally we derive the scale below which a self-interaction would take over from the cascade: for $1\,\mathrm{cm^2 g^{-1}}$ it is 0.046 pc. Along the way we find that the four-thirds law holds for the haloes as a population (slope -1.37) but not for every sample, and that the halo-particle rate in Illustris-1-Dark, $2.6 \times 10^{-7}\,\mathrm{m^2 s^{-3}}$, lies within the range that the rotation curves allow.

The paper is organized as follows. Section 2 derives the rate of the energy cascade from the cosmic energy equation and measures it in the Virgo and Illustris-1-Dark simulations. Section 3 presents the scaling laws of a constant cascade and tests them against simulated haloes and rotation curves. Section 4 extends the cascade to the particle scale and examines the robustness of the result. Section 5 discusses self-interacting dark matter, and Section 6 summarizes the conclusions.

2 THE RATE OF ENERGY CASCADE

2.1 Cosmic energy equation

For a self-gravitating system in an expanding background the specific peculiar kinetic energy K_p and the specific potential energy P_y obey the cosmic energy equation [25, 26]

$$\frac{\partial E_y}{\partial t} + H\left(2K_p + P_y\right) = 0, \qquad E_y = K_p + P_y, \tag{1}$$

with $H = \dot{a}/a$. Look for solutions in which both energies grow in proportion to time, $K_p = -\varepsilon_u t$ and $P_y = \alpha\,\varepsilon_u t$. With Ht constant,

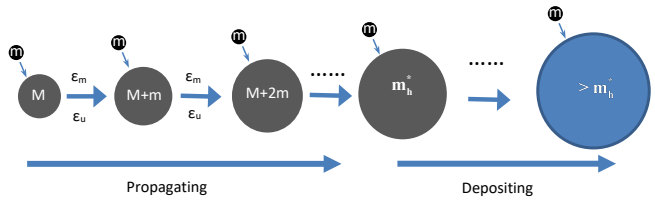


Figure 1. Inverse mass and energy cascade by mergers. A halo of mass M that absorbs a particle of mass m moves to the mass $M+m$, carrying mass and energy to larger scales. Below a characteristic mass m_h^* (the propagation range) haloes only pass mass and energy on, so the rates ε_m and ε_u do not depend on M; above m_h^* the cascaded mass is used to grow haloes. Schematic of the author.

Eq. (1) gives $(\alpha-1)+Ht(\alpha-2)=0$, so

$$\alpha = \frac{1+2Ht}{1+Ht} = \begin{cases} 7/5 & \text{matter era } (Ht = 2/3), \\ 4/3 & \text{radiation era } (Ht = 1/2). \end{cases} \tag{2}$$

The constant ε_u is the rate at which kinetic energy is cascaded; it is negative because the cascade is inverse. The total energy $E_y = K_p + P_y = (\alpha-1)\varepsilon_u t$ decreases with time. By Eq. (1) the loss is the work done by the cosmic expansion, $-H(2K_p + P_y)$, which equals $(\alpha-1)\varepsilon_u$ for the linear solution; it is not a dissipation of energy.

With $K_p = \frac{3}{2}u^2$ for a one-dimensional velocity dispersion u,

$$\varepsilon_u = -\frac{K_p}{t} = -\frac{3}{2}\frac{u_0^2}{t_0} = -4.58 \times 10^{-7}\ \mathrm{m^2\,s^{-3}}, \tag{3}$$

for the Virgo SCDM simulation [27, 28] with $u_0 = 354.6\,\mathrm{km\,s^{-1}}$ at the present time $t_0 = 2/(3H_0) = 13.04\,\mathrm{Gyr}$ ($h = 0.5$).

Strictly, ε_u is the rate at which the kinetic energy of halo particles changes with time. Writing the velocity of a halo particle as the mean velocity of its halo plus a virial part, the mean specific virial energy of halo particles grows as $\langle K_{pv}\rangle = |\varepsilon_u| t$ [20]. Equation (3) uses instead the kinetic energy K_p of all particles, in haloes and outside them, which also contains the bulk motion of haloes and the motion of particles that belong to no halo. It is a convenient estimate, given directly by the energy equation; it is not constant in time, but approaches the halo-particle rate as halo particles come to dominate the mass.

2.2 What the simulations show

Figure 2(a) shows the energies of the Virgo run. Between $a = 0.25$ and $a = 1$ they grow as $K_p \propto a^{1.26}$ and $-P_y \propto a^{1.33}$, slightly slower than the $a^{3/2} \propto t$ of a strictly constant rate, and the ratio $-P_y/K_p$ rises from 1.40 at $a = 1/3$ to 1.47 at $a = 1$, against 7/5 from Eq. (2). Figure 2(b) compares the two estimates. In the ΛCDM simulation Illustris-1-Dark [29] the all-particle estimate K_p/t falls from 7.9×10^{-7} at $z = 12$ to $3.3 \times 10^{-7}\,\mathrm{m^2 s^{-3}}$ at $z = 0$, whereas the halo-particle rate $\langle K_{pv}\rangle/t$ stays at $(2.6 \pm 0.1) \times 10^{-7}\,\mathrm{m^2 s^{-3}}$ for $a \geqslant 1/3$ ($\langle K_{pv}\rangle \propto t^{0.99}$). The all-particle estimate varies because the share of halo particles varies: the fraction of mass in haloes rises from 0.2 per cent at $z = 12$ to 55 per cent at $z = 0$, and the ratio $K_p/\langle K_{pv}\rangle$ falls from 3.8 at $z = 8$ to 1.35 at $z = 0$ (Fig. 2c). The part of K_p that is not virial motion follows the large-scale flow and stops growing once dark energy dominates (the bulk energy of haloes falls as $t^{-0.41}$ between $z = 0.3$ and 0), so K_p/t falls fastest at late times. In the Virgo run the halo-particle rate is 3.0–$3.7 \times 10^{-7}\,\mathrm{m^2 s^{-3}}$ for $a \geqslant 1/4$ and 3.0×10^{-7} at $a = 1$, where 61 per cent of the mass is in haloes and Eq. (3) is 1.53 times larger. The all-particle estimate thus exceeds the halo-particle rate by a factor that shrinks as haloes grow. We

keep Eq. (3) as the reference value and carry the range from the halo-particle rates to the rotation curves in Sections 3 and 4; at the Illustris halo-particle rate the particle mass of Section 4 would be $0.85\times10^{12}\,\mathrm{GeV}$.

The mass cascade that accompanies the energy cascade is measured in the same way from the halo mass function of Illustris-1-Dark: below m_h^* its rate is independent of halo mass, about $0.022\,M_\odot\,\mathrm{s}^{-1}$ between $z=0.3$ and $z=0$ (the measurement is described in Xu 23).

3 SCALING LAWS OF A CONSTANT ENERGY CASCADE

3.1 Two-thirds, five-thirds and four-thirds laws

If the kinetic energy v_r^2 on scale r is passed on in one turnover time r/v_r at the constant rate ε_u,

$$\frac{2v_r^2}{r/v_r}=-\lambda_u\varepsilon_u, \tag{4}$$

with λ_u a number of order unity, then $v_r^2\propto(-\varepsilon_u r)^{2/3}$: the two-thirds law, which the pairwise velocities of the Virgo simulation follow on small scales once the reduced dispersion is used [30]. With the virial theorem $Gm_r/r\propto v_r^2$, the mass m_r and the mean density ρ_r within r are

$$m_r=\alpha_r\,|\varepsilon_u|^{2/3}G^{-1}r^{5/3},\qquad \rho_r=\beta_r\,|\varepsilon_u|^{2/3}G^{-1}r^{-4/3},\qquad \alpha_r=\tfrac{4\pi}{3}\beta_r. \tag{5}$$

Only ε_u, G and r enter, so the same laws follow from dimensional analysis. The largest scale of the propagation range is $r_l=u_0^3/|\varepsilon_u|=\frac{2}{3}u_0t_0=3.15\,\mathrm{Mpc}$ ($1.58\,h^{-1}\,\mathrm{Mpc}$).

3.2 Haloes in Illustris-1-Dark

Within 16 bins of halo mass at $z=0$, 4, 8 and 12, the enclosed mass of Illustris-1-Dark haloes between $0.1r_s$ and r_s (r_s: radius where the logarithmic density slope is -2) grows as $m_r\propto r^s$ with a median $s=1.61$ (range 1.49–1.96) against $5/3$. This is a consistency check rather than a test: an NFW profile [31] has the mean slope 1.64 over the same interval. The more demanding comparison is across haloes of very different size.

3.3 Galaxy rotation curves

Decomposed rotation curves give, for each galaxy, a halo scale radius r_s and the mean density ρ_s within it. We use the fits of three samples with two halo models: SPARC, 153 galaxies with NFW and pseudo-isothermal (pISO) fits [32, 33]; the DiskMass Survey, 30 galaxies, NFW and pISO [34]; and 43 galaxies with NFW fits compiled by Sofue [35]. The SPARC points were recomputed from the published fit parameters ($\rho_s=f(c)M_{200}/(\frac{4}{3}\pi r_s^3)$ for NFW, $3\rho_0(1-\pi/4)$ for pISO [36]) and agree with the plotted values to machine precision for all 153 galaxies.

Figure 3 shows the 409 haloes. They span 3.9 decades in r_s and 5.3 in ρ_s, about 6.9 decades in halo mass $\rho_s r_s^3$. Together they follow Eq. (5): the least-squares slope of $\log\rho_s$ on $\log r_s$ is -1.37, with a bootstrap 95 per cent interval -1.44 to -1.30 that contains $-4/3$. Table 1 and Fig. 4 give the samples separately. Their slopes range from -1.07 (SPARC, NFW) to -1.77 (DMS, pISO), and only the SPARC pISO sample contains $-4/3$ in its interval. The SPARC NFW points alone are closer to a constant surface density $\rho_s r_s$ (slope -1), the relation found for cored fits by Donato *et al.* [37]. Part of the difference between samples is in the halo models: the pooled NFW points give -1.28 and the pooled pISO points -1.44. The four-thirds law is thus a fair description of the population of haloes as a whole, not of every sample.

Table 1. The four-thirds law in rotation-curve samples: least-squares slope of $\log\rho_s$ on $\log r_s$ with bootstrap 95 per cent interval, median β_r (Eq. 5 with ε_u of Eq. 3) and scatter about the law (dex).

sample	N	slope (95%)	β_r	scatter
SPARC, NFW	153	-1.07 (-1.24, -0.90)	0.67	0.37
SPARC, pISO	153	-1.34 (-1.47, -1.20)	0.74	0.39
DMS, NFW	30	-1.69 (-1.90, -1.58)	1.14	0.20
DMS, pISO	30	-1.77 (-1.91, -1.64)	2.13	0.22
Sofue, NFW	43	-1.56 (-1.68, -1.44)	1.69	0.14
all	409	-1.37 (-1.44, -1.30)	0.86	–

The normalisation β_r of Eq. (5) at ε_u of Eq. (3) has a median of 0.86 over all haloes and 0.67–2.13 by sample, with a scatter of 0.14–0.39 dex about the law (Table 1). Read the other way, with $\beta_r=1.26$ (the line in Fig. 3) the samples imply $|\varepsilon_u|$ between 1.8×10^{-7} and $1.0\times10^{-6}\,\mathrm{m^2s^{-3}}$. Together with the simulations, the data allow

$$1.8\times10^{-7}\leqslant|\varepsilon_u|\leqslant1.0\times10^{-6}\ \mathrm{m^2\,s^{-3}}. \tag{6}$$

4 EXTENSION TO THE PARTICLE SCALE

4.1 Three conditions for two particles

Without viscosity, and with gravity as the only interaction, nothing in the dark matter fluid ends the cascade above the scale where the particles' wave nature matters. We assume that the rate ε_u and the two-thirds law hold down to that scale, and consider two particles of mass m_X a distance l_X apart in their centre-of-mass frame, each moving with speed V_X:

$$m_X V_X\frac{l_X}{2}=\hbar \qquad \text{(uncertainty principle)}, \tag{7}$$

$$\frac{2V_X^3}{l_X}=-\lambda_u\varepsilon_u \qquad \text{(constant energy flux, Eq. 4)}, \tag{8}$$

$$\frac{Gm_X}{l_X}=2V_X^2 \qquad \text{(virial theorem)}. \tag{9}$$

Eliminating l_X and m_X gives $V_X^9=G\hbar\lambda_u^2\varepsilon_u^2/4$, and then

$$m_X=\left(\frac{256\lambda_u|\varepsilon_u|\hbar^5}{G^4}\right)^{1/9},\qquad l_X=\left(\frac{2G\hbar}{\lambda_u|\varepsilon_u|}\right)^{1/3},\qquad t_X=\frac{l_X}{V_X}=\left(\frac{32G^2\hbar^2}{\lambda_u^5|\varepsilon_u|^5}\right)^{1/9}. \tag{10}$$

Substituting Eq. (10) back into Eqs. (7)–(9) leaves residuals below 10^{-14}. Table 2 lists the values for $\lambda_u=1$ and ε_u of Eq. (3), with CODATA constants. Dimensional analysis alone gives the same combinations without the numerical factors (for example $(|\varepsilon_u|\hbar^5/G^4)^{1/9}=0.49\times10^{12}\,\mathrm{GeV}$); the factors come from Eqs. (7)–(9).

4.2 How robust are the numbers?

The mass depends on the rate only through $|\varepsilon_u|^{1/9}$ (Fig. 5). The whole range of Eq. (6) moves m_X between 0.82 and $0.99\times10^{12}\,\mathrm{GeV}$,

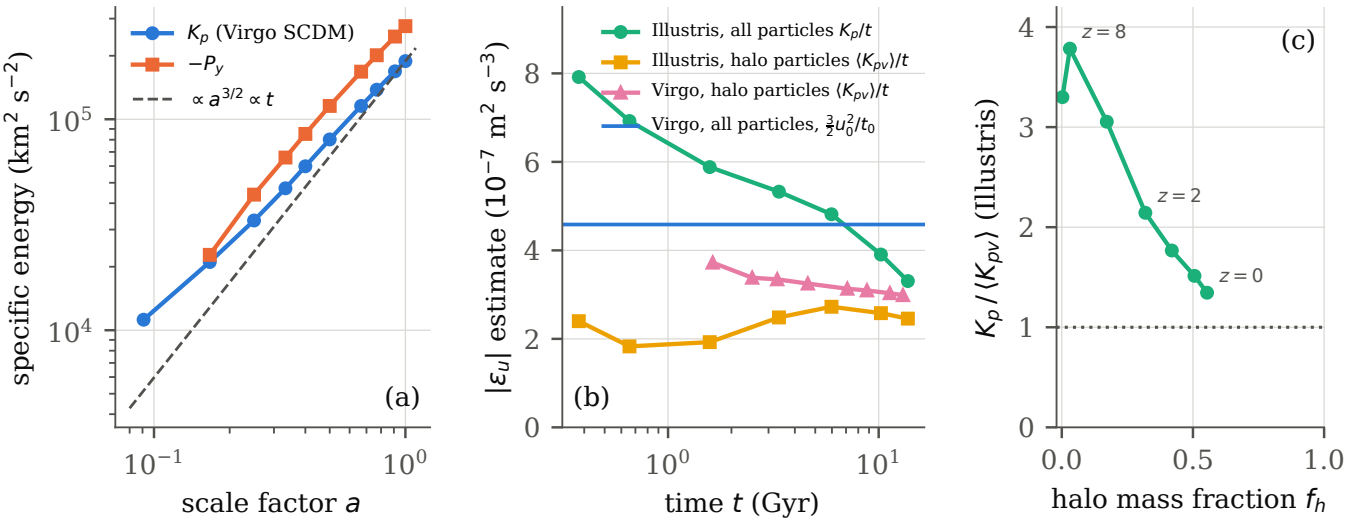


Figure 2. (a) Specific kinetic energy K_p and potential energy $-P_y$ of the Virgo SCDM simulation against the scale factor, with the slope $a^{3/2} \propto t$ of a constant rate of cascade. (b) Estimates of the rate of cascade: K_p/t of all particles and $\langle K_{pv}\rangle/t$ of halo particles in Illustris-1-Dark at its seven snapshots ($z = 12$ to 0), $\langle K_{pv}\rangle/t$ of halo particles in Virgo SCDM, and the all-particle Virgo value of Eq. (3). (c) Ratio of the all-particle to the halo-particle specific kinetic energy in Illustris-1-Dark against the fraction of mass in haloes. Data: plotted arrays of the author's figure files.

Table 2. Particle-scale quantities from Eq. (10) ($\lambda_u = 1$, $\varepsilon_u = -4.58 \times 10^{-7}\,\mathrm{m^2 s^{-3}}$) and their range for $|\varepsilon_u|$ in Eq. (6). The exponent gives $d\ln q/d\ln|\varepsilon_u|$.

quantity	value	range	exponent
m_X	0.91×10^{12} GeV ($= 1.62 \times 10^{-15}$ kg)	0.82–0.99 $\times 10^{12}$ GeV	1/9
l_X	3.13×10^{-13} m	2.4–4.3 $\times 10^{-13}$ m	−1/3
t_X	7.53×10^{-7} s	4.9–12.8 $\times 10^{-7}$ s	−5/9
V_X	4.16×10^{-7} m s^{-1}	3.37–4.95 $\times 10^{-7}$	2/9
$a_X = 2V_X^2/l_X$	1.10 m s^{-2}		7/9
$E_X = \hbar/t_X$	0.87 neV	0.52–1.35 neV	5/9
$\rho_X = m_X/l_X^3$	5.28×10^{22} kg m^{-3}	1.8–12.7 $\times 10^{22}$	10/9
$P_X = m_X a_X/l_X^2$	1.82×10^{10} Pa		14/9

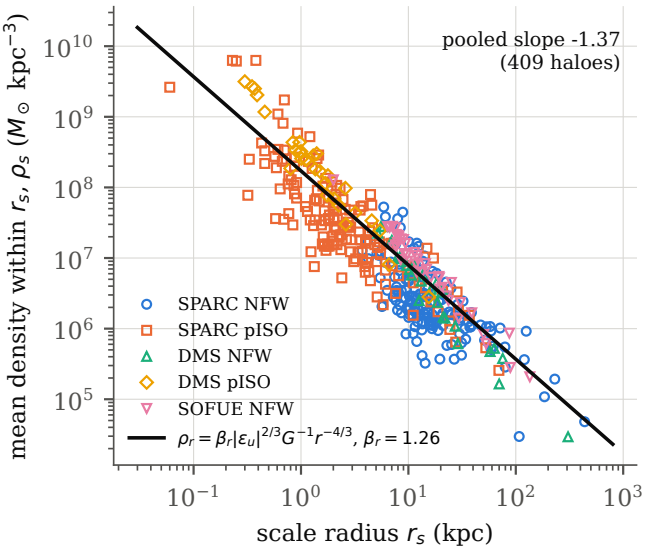


Figure 3. Mean halo density ρ_s within the scale radius r_s from rotation-curve decompositions: SPARC (NFW and pISO fits), DiskMass Survey (DMS) and Sofue's compilation. The line is the four-thirds law, Eq. (5), with ε_u of Eq. (3) and $\beta_r = 1.26$. Data: author's figure file; the SPARC points are recomputed from Li *et al.* [33].

and λ_u between 1/2 and 2 moves it between 0.84 and 0.98 $\times$ 10^{12} GeV; both together give 0.76–1.07 $\times$ 10^{12} GeV. Even a rate a thousand times smaller than the data allow would only halve the mass. Other quantities are more sensitive: $l_X \propto |\varepsilon_u|^{-1/3}$, $t_X \propto |\varepsilon_u|^{-5/9}$ and the energy scale $\hbar/t_X \propto |\varepsilon_u|^{5/9}$ change by factors of 1.8, 2.6 and 2.6 over the same range.

What the numbers rest on is not the value of ε_u but the assumption behind Eqs. (7)–(9): that the inverse cascade of collisionless haloes continues, with the same rate, over about 30 decades in length from galactic scales to 10^{-13} m. Neither simulations nor rotation curves reach below about 50 pc, so the assumption is untested. If it fails because some interaction other than gravity becomes important (Section 5), Eq. (10) no longer applies.

4.3 Comparison with other routes

The same mass scale appears in a different argument: 10^{12} GeV is the mass for which the free-streaming mass equals the particle mass, so that the smallest bound structures are made of a few particles [23]. The two arguments use different physics (free streaming in the early universe in one, the late-time energy cascade in the other), and their agreement is the main reason to take the scale seriously. Superheavy dark matter of this mass can be produced gravitationally at the end

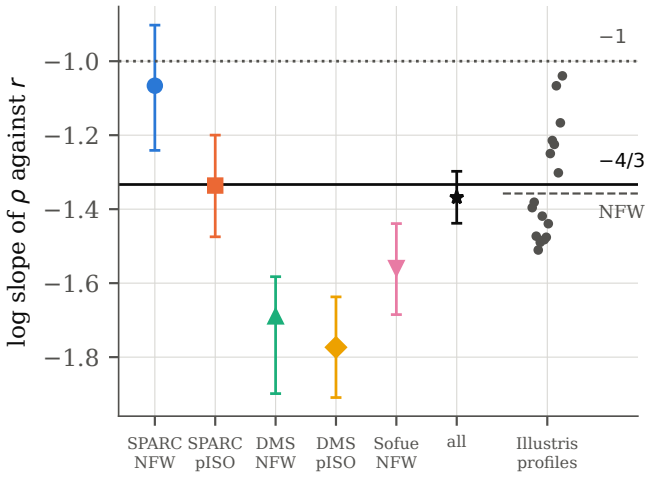


Figure 4. Logarithmic slope of $\rho_s(r_s)$ for each rotation-curve sample and for all 409 haloes (least squares, bootstrap 95 per cent intervals), and slopes of the enclosed density within 0.1–1 r_s of the 16 Illustris-1-Dark halo bins (grey dots). Lines: the four-thirds law ($-4/3$), a constant surface density (-1) and the NFW profile over the same radial interval.

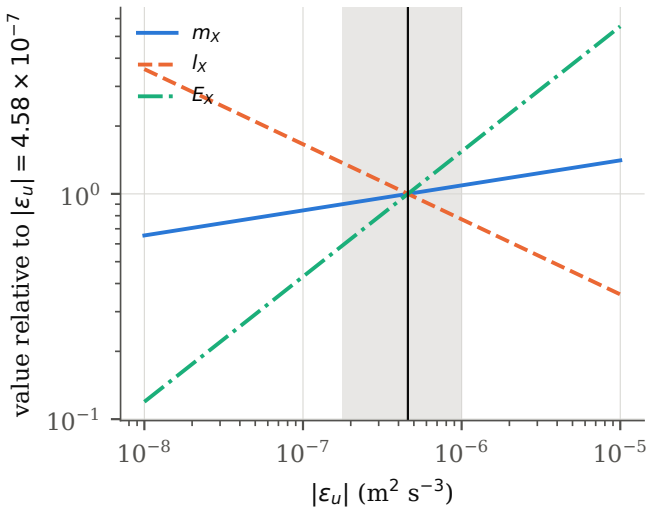


Figure 5. Particle mass m_X, size l_X and energy scale $E_X = \hbar/t_X$ from Eq. (10) against $|\varepsilon_u|$, relative to their values at $|\varepsilon_u| = 4.58 \times 10^{-7}\,\mathrm{m^2 s^{-3}}$ (vertical line). Shaded: the range allowed by the data, Eq. (6).

of inflation [7, 8]; the production and the relic abundance of this particle are treated in Xu [23].

Two derived numbers describe the particles' interactions with each other. The product $l_X^2 V_X = 4.1 \times 10^{-32}\,\mathrm{m^3 s^{-1}}$ is of the order of the thermally averaged cross section of a thermal relic, $3 \times 10^{-32}\,\mathrm{m^3 s^{-1}}$; the ratio $l_X^2/m_X = 6.1 \times 10^{-11}\,\mathrm{m^2 kg^{-1}}$ is about 6×10^{-10} times the $1\,\mathrm{cm^2 g^{-1}}$ discussed for self-interacting dark matter, so the particles are collisionless on all astrophysical scales, as assumed. The mean separation of such particles is 131 m in the solar neighbourhood ($7.2 \times 10^{-22}\,\mathrm{kg\,m^{-3}}$) and about 10 km on cosmic average.

5 SELF-INTERACTING DARK MATTER

If dark matter particles scatter with a cross section per unit mass σ/m, a third constant enters on small scales. Combined with ε_u and G it defines a length, time, mass and density; for a halo on the four-thirds law the length

$$r_\eta = \beta_r^3 \frac{\varepsilon_u^2 (\sigma/m)^3}{G^3} \tag{11}$$

is where $\rho_r(\sigma/m)r = 1$, i.e. where a particle scatters about once per crossing of the scale. Below r_η self-interaction, not the cascade, controls the structure, and the four-thirds law should end at the density $\rho_\eta = \varepsilon_u^{-2} G^3 (\sigma/m)^{-4} \beta_r^{-3}$. Table 3 gives the values for $\beta_r = 1.26$. For $\sigma/m = 1\,\mathrm{cm^2 g^{-1}}$ the cascade is dominated by scattering only below 0.046 pc. At 1 kpc a four-thirds halo ($0.17\,M_\odot\,\mathrm{pc^{-3}}$) has a scattering depth of only 0.04 per crossing, so the kiloparsec cores attributed to self-interaction must build up over many crossing times rather than within one cascade step. Self-interaction strong enough to alter galactic cores would therefore leave the rate ε_u on galactic scales unchanged but stop the cascade far above the particle scale, and Eq. (10) would not apply.

Table 3. Scales where self-interaction takes over from the cascade, Eq. (11), for $\beta_r = 1.26$ and ε_u of Eq. (3).

σ/m ($\mathrm{cm^2 g^{-1}}$)	r_η	ρ_η ($M_\odot\,\mathrm{pc^{-3}}$)
0.1	9.5 AU	1.0×10^9
1	0.046 pc (9453 AU)	1.0×10^5

6 CONCLUSIONS

- The rate of the inverse energy cascade of dark matter is set by the kinetic energy of halo particles: $\langle K_{py} \rangle / t = (2.6 \pm 0.1) \times 10^{-7}\,\mathrm{m^2 s^{-3}}$ in Illustris-1-Dark and 3.0×10^{-7} in Virgo today. The kinetic energy of all particles gives the convenient estimate $\varepsilon_u \approx -4.58 \times 10^{-7}\,\mathrm{m^2 s^{-3}}$ (Virgo) used here; in Illustris-1-Dark it falls from 7.9 to 3.3×10^{-7} and approaches the halo-particle rate as the halo mass fraction grows.
- The four-thirds law $\rho_r \propto |\varepsilon_u|^{2/3} G^{-1} r^{-4/3}$ describes 409 rotation-curve haloes as a population (slope -1.37, interval -1.44 to -1.30); single samples range from -1.07 to -1.77, and their normalisations allow $|\varepsilon_u|$ between 1.8×10^{-7} and $1.0 \times 10^{-6}\,\mathrm{m^2 s^{-3}}$.
- If the cascade continues to the quantum scale with gravity as the only interaction, the particles have $m_X = 0.91 \times 10^{12}$ GeV (range 0.76–1.07 $\times 10^{12}$ GeV for all allowed rates and $\lambda_u = 1/2$–2), $l_X = 3.13 \times 10^{-13}$ m and an energy scale of 0.87 neV. The mass agrees with the critical mass from free streaming.
- A self-interaction of $1\,\mathrm{cm^2 g^{-1}}$ would take over from the cascade only below 0.046 pc.

The weakest link is the extrapolation of the cascade over some 30 decades in scale. Its consequences that can be tested are the constancy of ε_u in simulations with smaller particle masses, the four-thirds law in dwarf galaxies of the smallest r_s, and the absence of any interaction of dark matter beyond gravity down to the scales probed.

DATA AVAILABILITY

The halo-based and correlation-based statistics of the Virgo simulation are on Zenodo [38, 39]. The plotted data of every figure, the SPARC fit tables used, and the scripts that compute every number of this paper are in the accompanying reproducibility package.

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